

A Photo Introduction to Kryptos

This is a quick introduction to the Kryptos encrypted sculpture at CIA headquarters in Virginia.

It is a guide to the pieces that make up Kryptos and it's history.

There is some information about the parts already solved at the end.



What is Kryptos

Kryptos is an encrypted sculpture by American artist Jim Sanborn located on the grounds of the Central Intelligence Agency (CIA) in Langley, Virginia. When the CIA was constructing a new headquarters building behind its original headquarters they wanted an outdoor art installation for the area between the two buildings. Since its dedication on November 3, 1990, there has been much speculation about the meaning of the encrypted messages it bears. Of the four messages (known as K1, K2, K3 and K4), three have been solved, with the fourth remaining one of the most famous unsolved codes in the world.

The name Kryptos comes from the Greek word for "hidden", and the theme of the sculpture is "intelligence gathering." The most prominent feature is a large vertical S-shaped copper screen resembling a scroll, or piece of paper emerging from a computer printer, covered with characters constituting encrypted text. The characters consist of the 26 letters of the standard Latin alphabet and question marks cut out of the copper. The sculpture gives the appearance there was once something else written underneath, then wiped away, and the new text written, a "palimpsest". A palimpsest is a manuscript page, either from a scroll or a book, from which the text has been either scraped or washed off so that the page can be used again. Palimpsest was a keyword used to solve K1.

There is more to Kryptos than just the sculpture

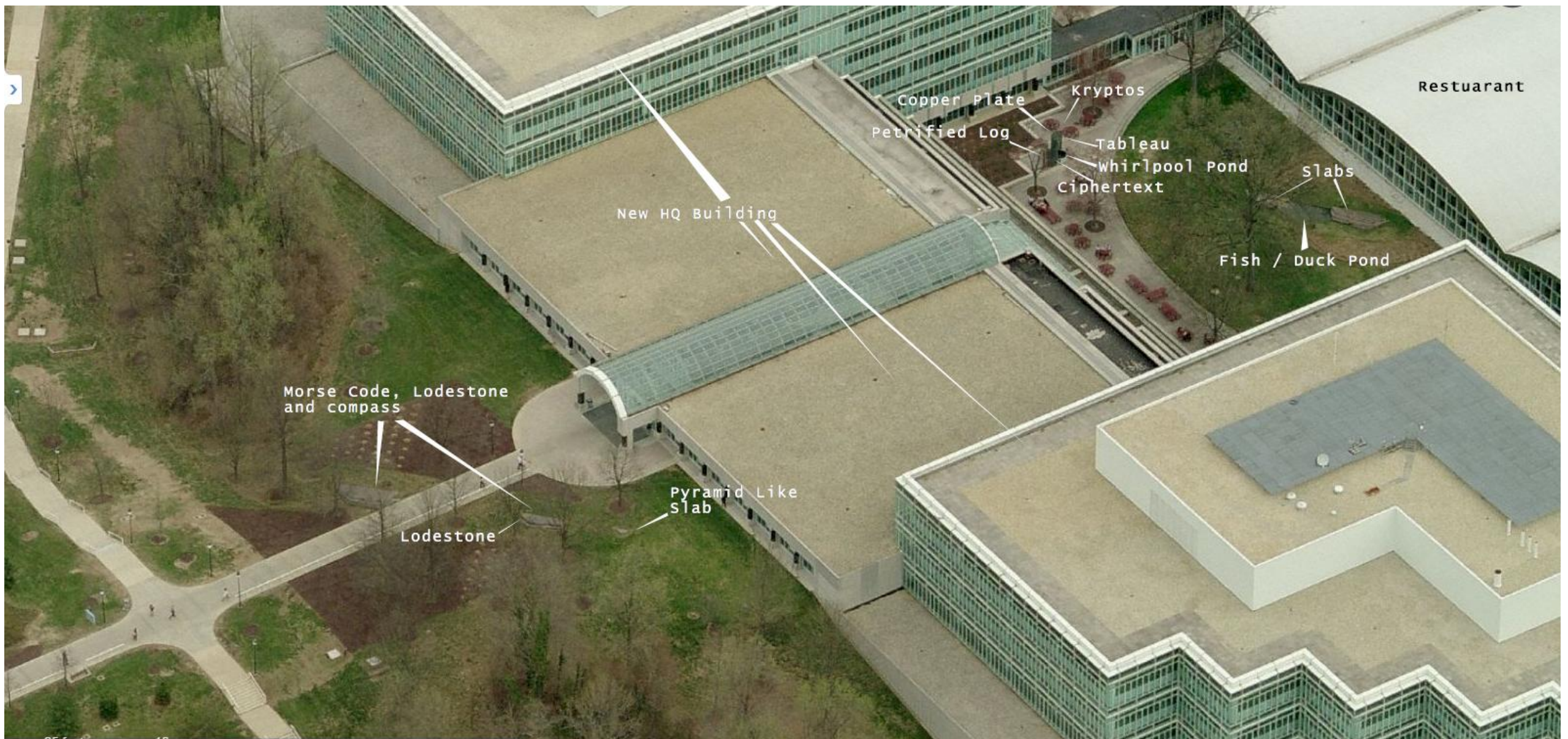
At the same time as the main sculpture was installed, sculptor Jim Sanborn also placed several other pieces around CIA grounds, such as several large granite slabs with sandwiched copper sheets outside the entrance to the New Headquarters Building. Several morse code messages are engraved in the copper, and one of the slabs has an engraved compass rose and a lodestone. Other elements of Sanborn's installation include a landscaped area, a fish pond, a reflecting pool, and several other seemingly unmarked slabs. There are some elements around CIA grounds the public has not seen. So far only the sculpture itself was needed to solve the first 3 parts. So far only Jim Sanborn knows the answer to K4.

On the front lawn in front of the entrance to the New HQ Building there are: Morse code on slabs, a lodestone, an engraved compass pointing to the lodestone and a pyramid shaped slab.

In the courtyard there are: The S shaped sculpture with petrified tree, whirlpool, fish pond, slabs with no apparant markings, 3 benches and a large letter J formed from plants that cuts into the restuarant building. Jim Sanborn designed only the semi circle landscaped area and it's features and the S shaped sculpture and pool and the boulder, slate, slabs. He did not design the zig zag walls or other parts of the courtyard.

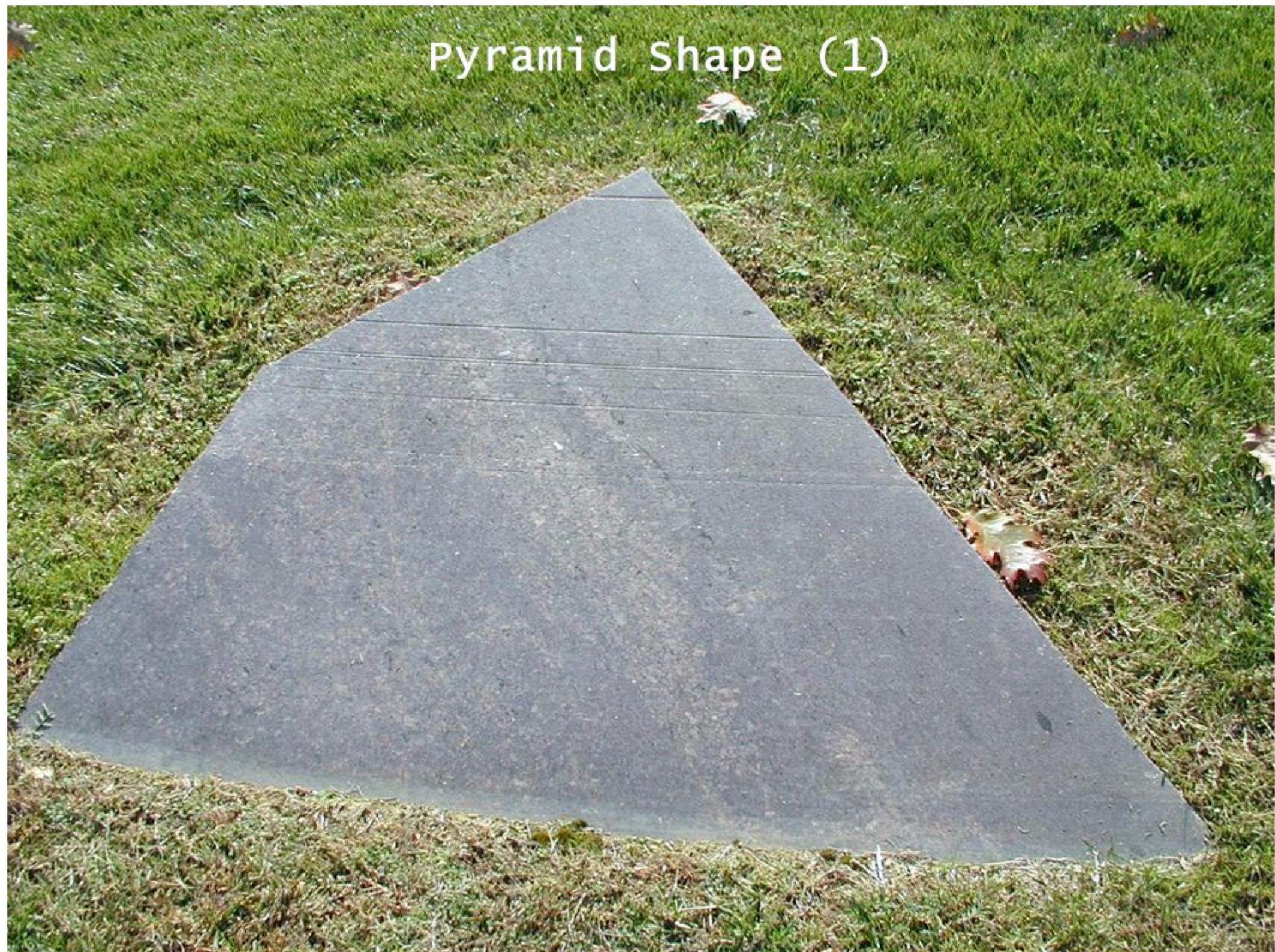
Jim Sanborn said you should look at the art installation as if you were flying over it. A copy of Jim Sanborns letter to CIA employees describing the installation is on the last page.

Photos



The front lawn





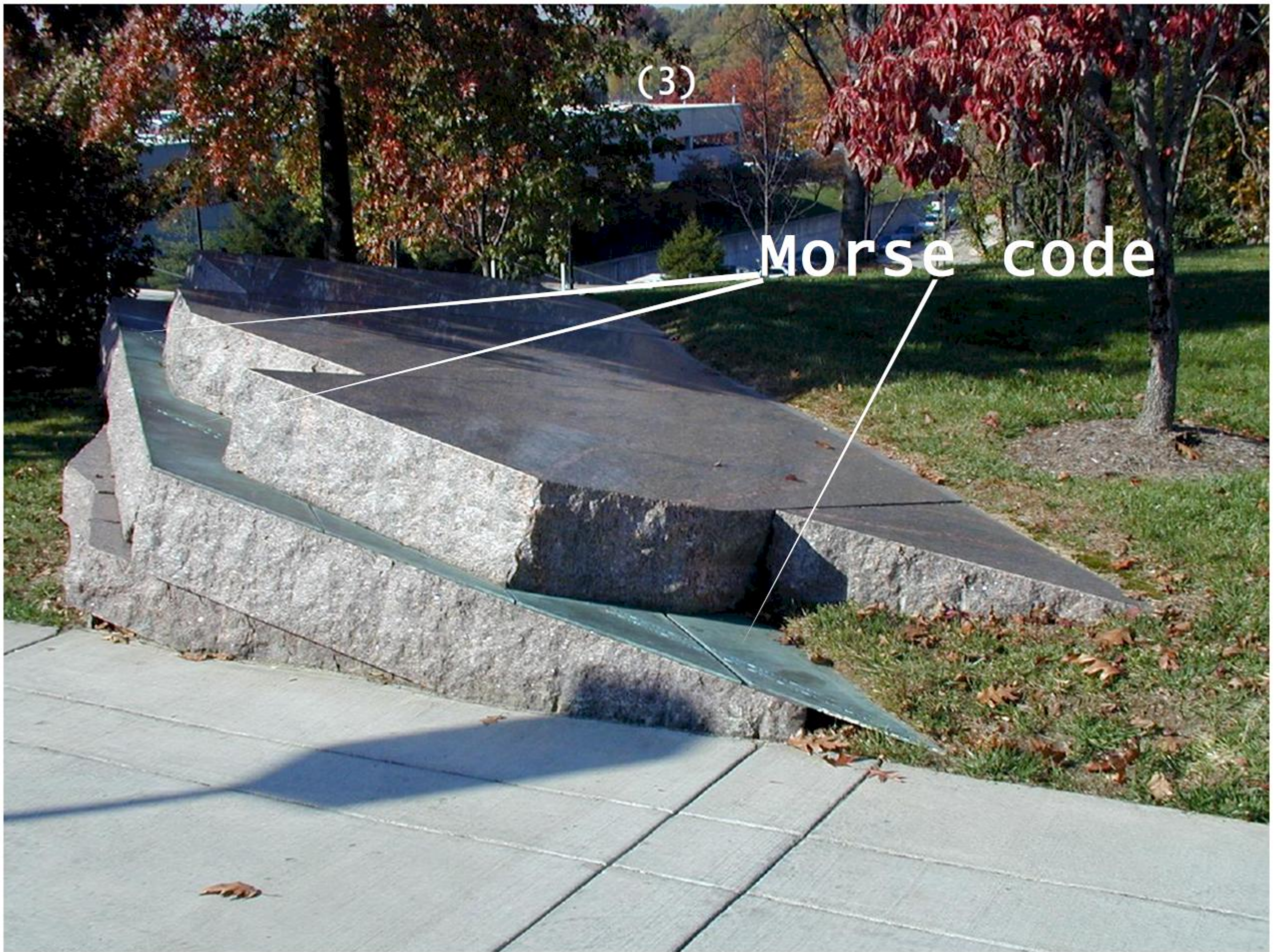
Some people have commented that it also looks like an arrow head. The letters K V W X Y feature arrow heads in their shapes





As shown in the photo the compass will not point North, it will always point to the lodestone. A lodestone, or loadstone, is a naturally magnetized piece of the mineral magnetite. They are naturally-occurring magnets, which can attract iron. The property of magnetism was first discovered through lodestones. Pieces of lodestone, suspended so they could turn, were the first magnetic compasses, and their importance to early navigation is indicated by the name lodestone, which in Middle English means 'course stone' or 'leading stone', from the now-obsolete meaning of lode as 'journey, way'. Lodestone is one of only two minerals that is found naturally magnetized; the other is pyrrhotite.

Lodestones were once believed to possess magical properties of various kinds. If one was ill, the stone should be held in the hands and shaken well. It was said to cure wounds, snakebites, weak eyes, headaches, and defective hearing. The possessor of the lodestone was supposed to be able to walk through reptiles in safety even when they were accompanied by "black death." Orpheus stated that "with this stone you can hear the voices of the gods and learn many wonderful things," that it had the property of unfolding the future, and if held close to the eyes, it would inspire with a divine spirit.



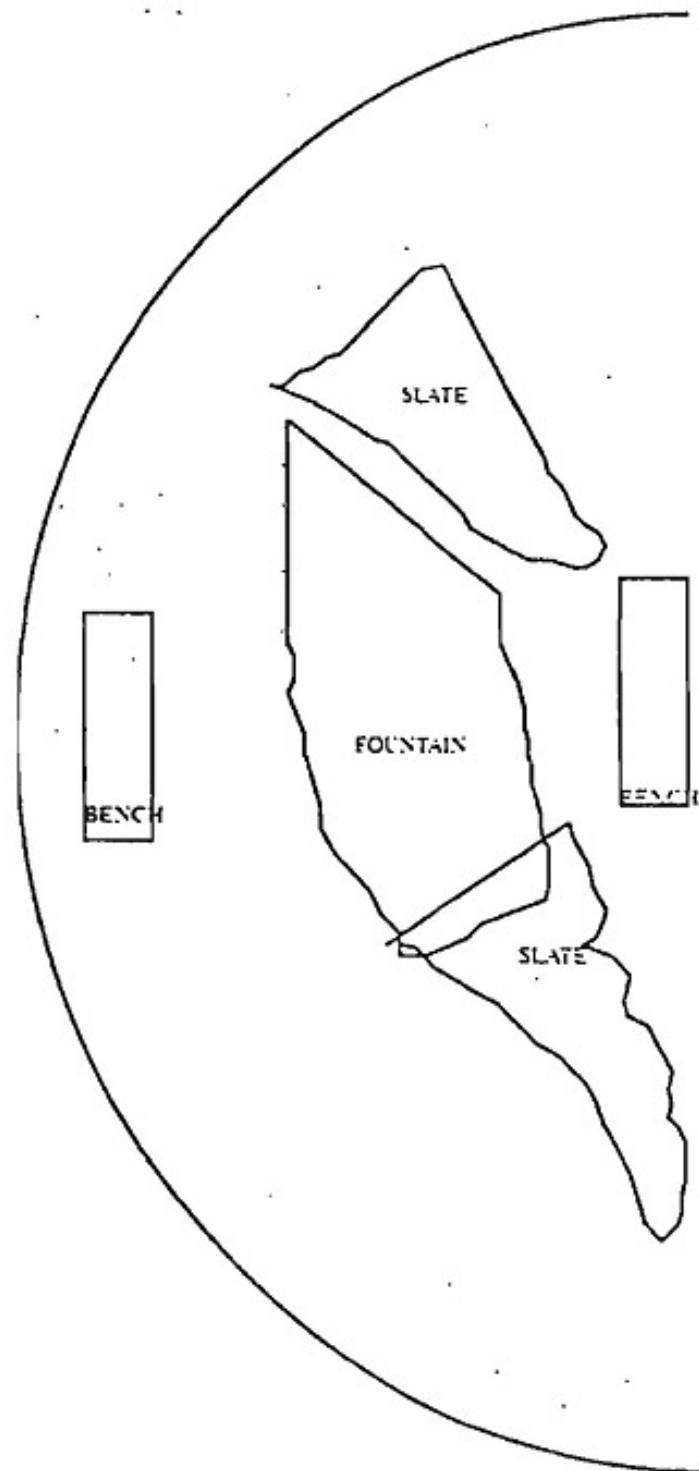
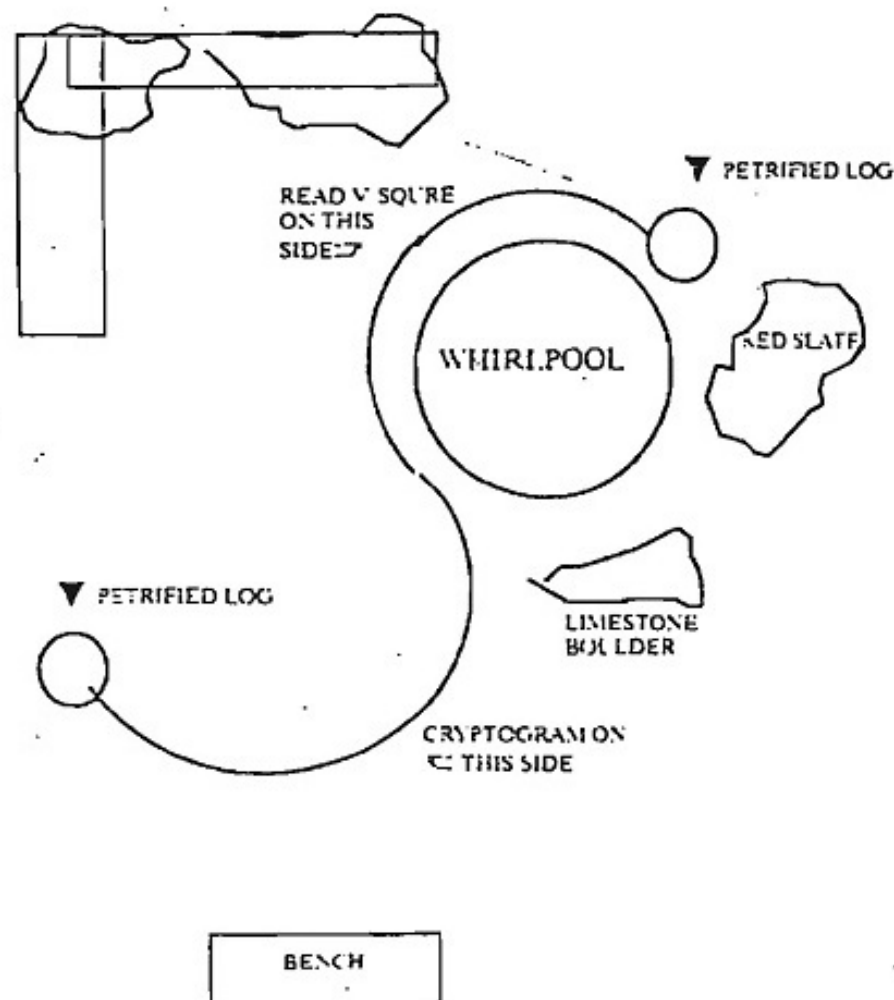
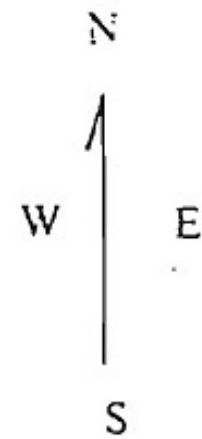
According to Jim Sanborn the slabs represent pages in a book, the copper sheets with morse code are the text on the pages.
It could represent a lost book, a discarded book, a buried book, a found book, a book of things that are buried or a famous book such as the Archimedes Palimpsest



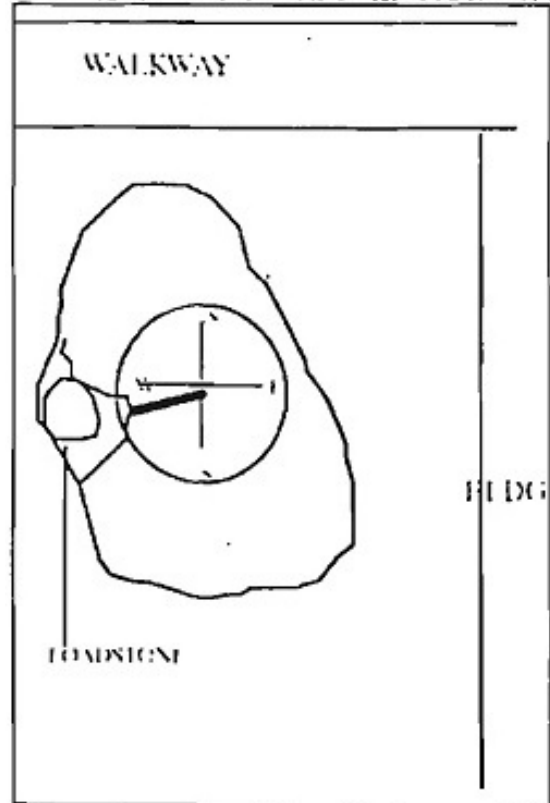
The Courtyard



(b) (3) - P.L. 86-36



OUTSIDE FRONT (WEST) ENTRANCE:



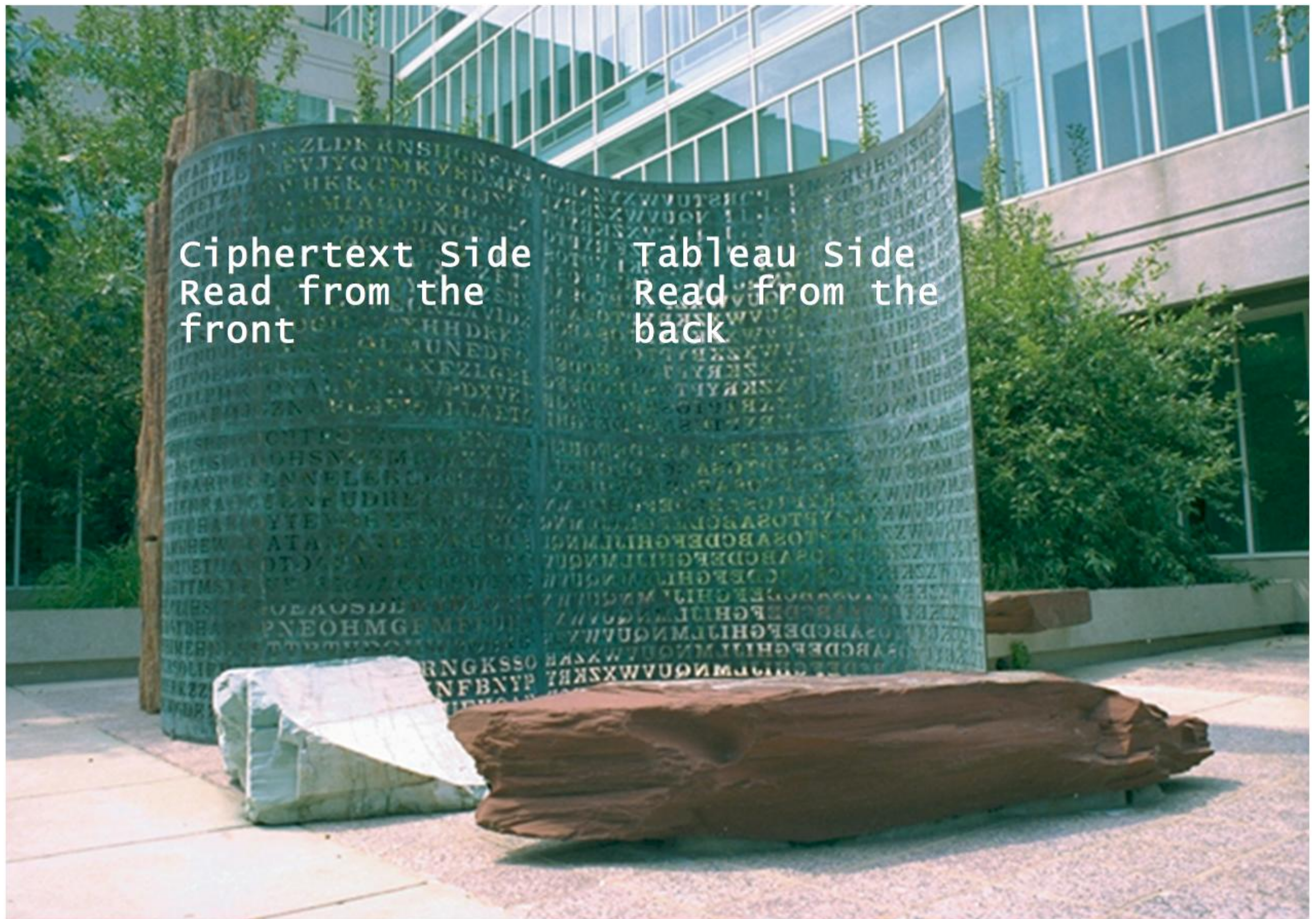
Note only the compass and lodestone are included in this drawing. No morse code. The morse code was not needed to solve the first 3 parts of Kryptos. When commenting in 2006 about his error in section K2, Jim Sanborn said that the answers to the first sections contain clues to the last section.

The sculpture



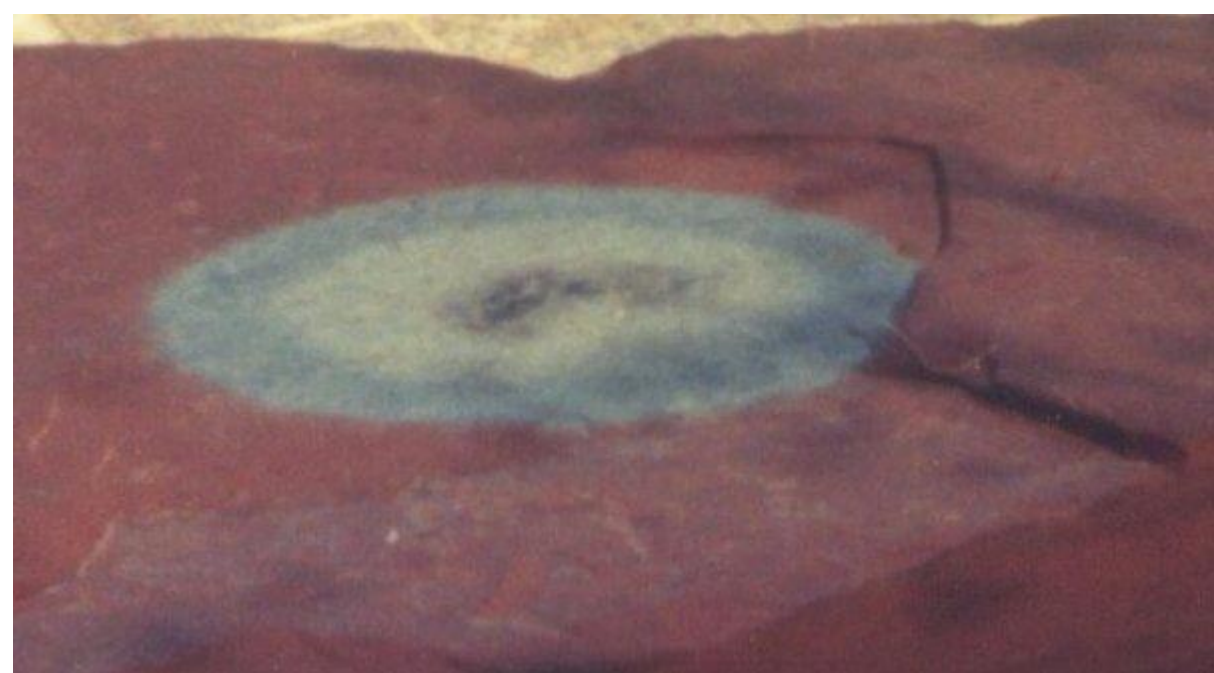
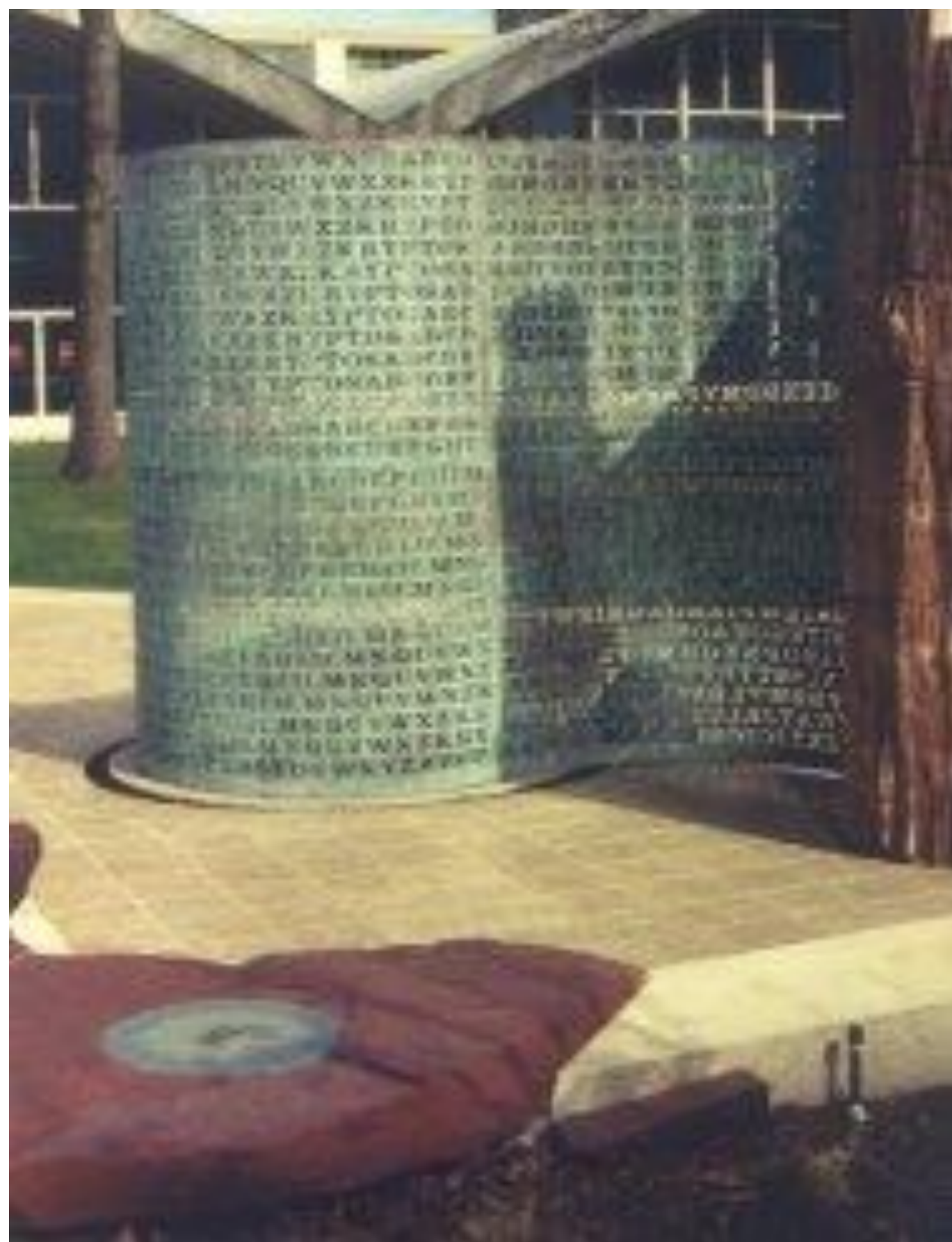
Ciphertext Side
Read from the
front

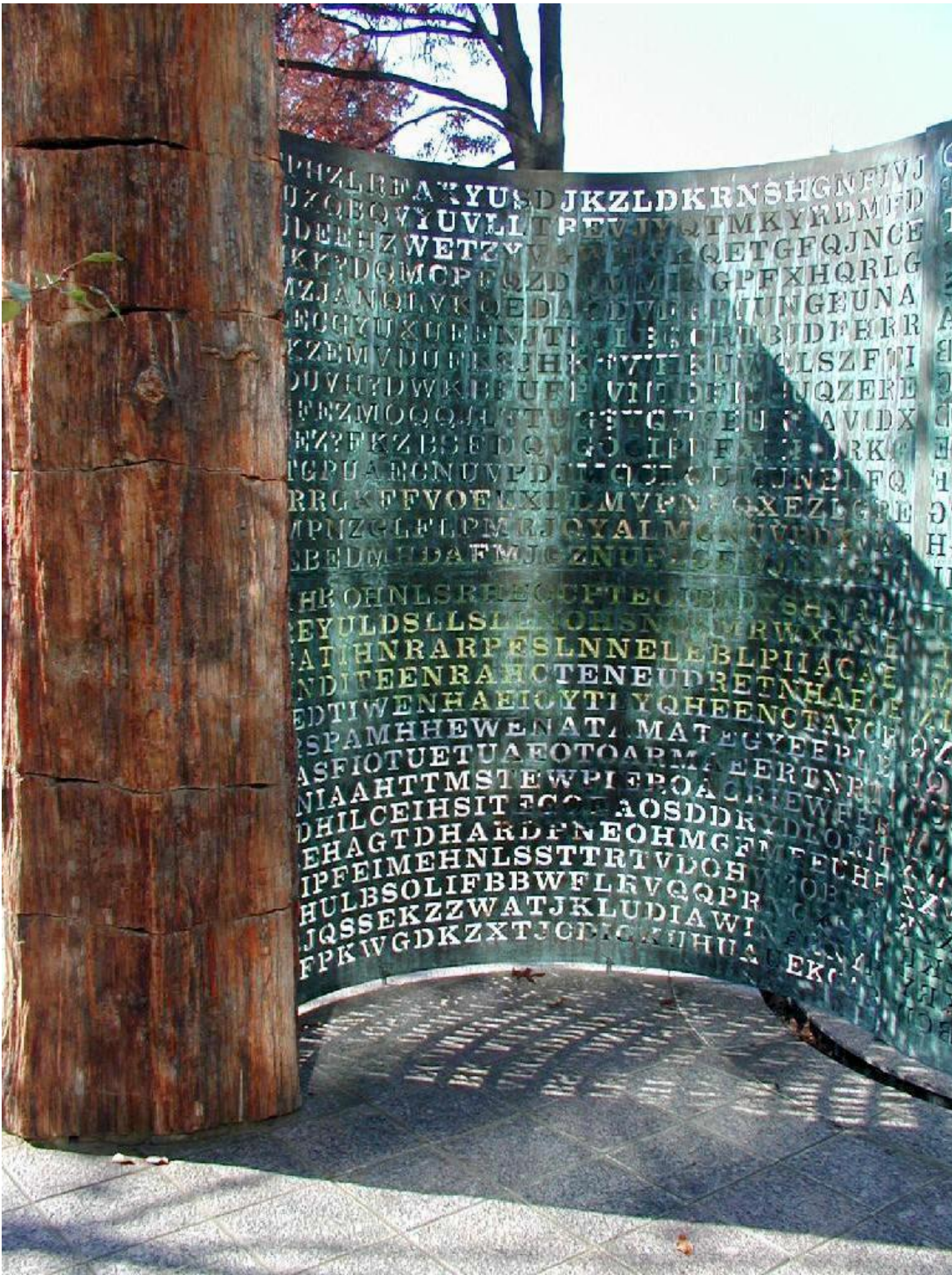
Tableau Side
Read from the
back





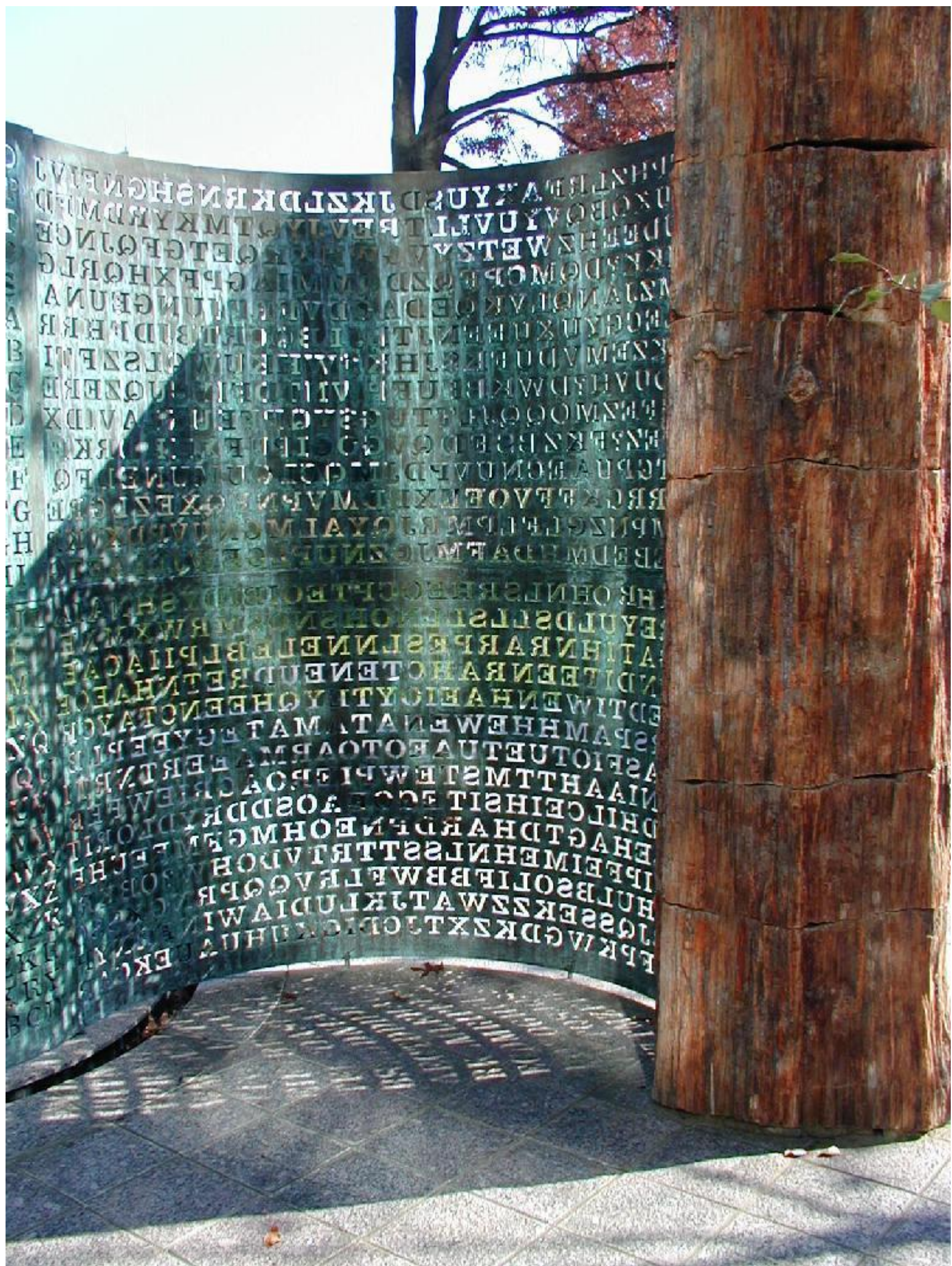






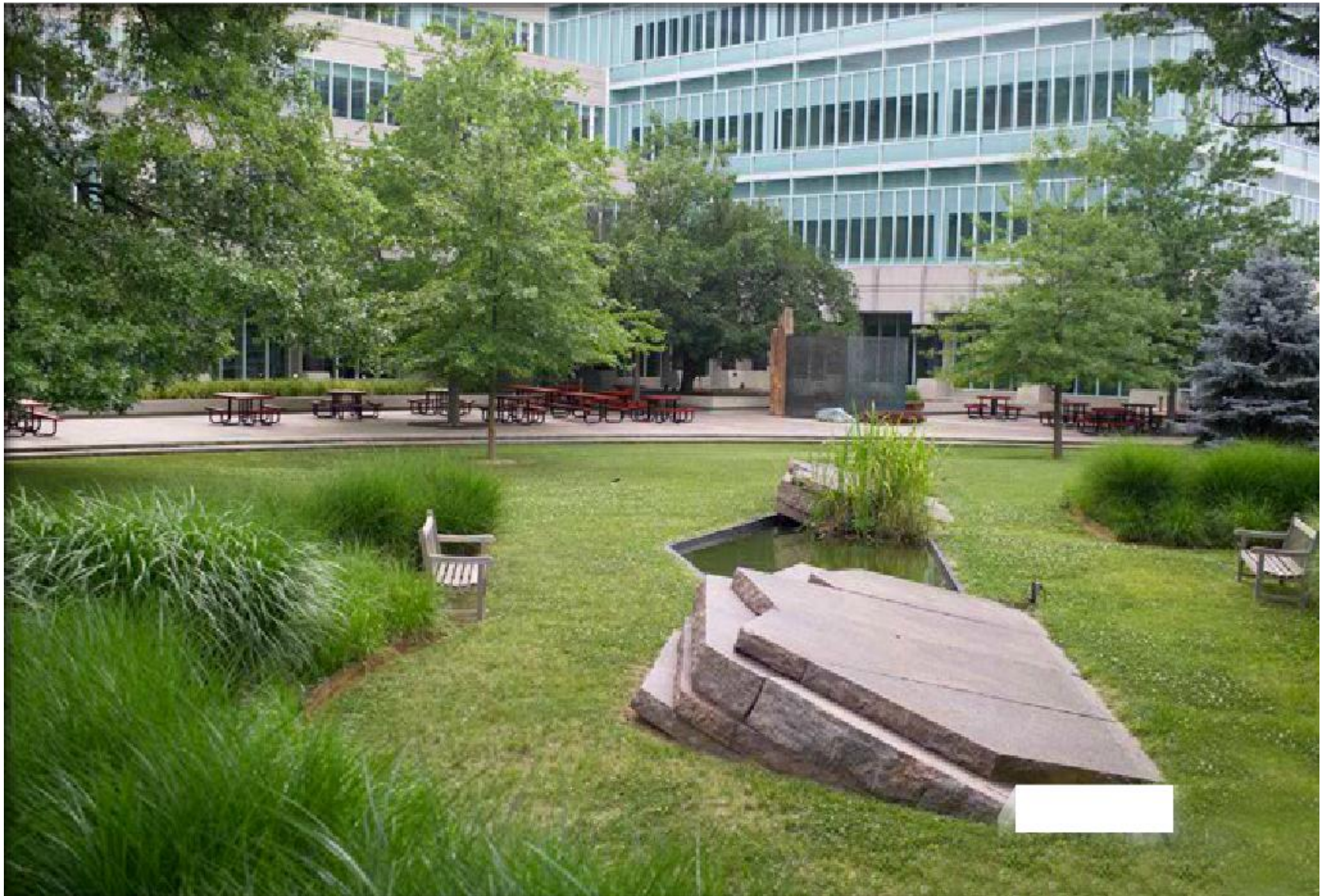
The cut outs in the petrified tree line up with ciphertext lines 3, 7, 11, 19, 23 (prime numbers)







The Fish Pond













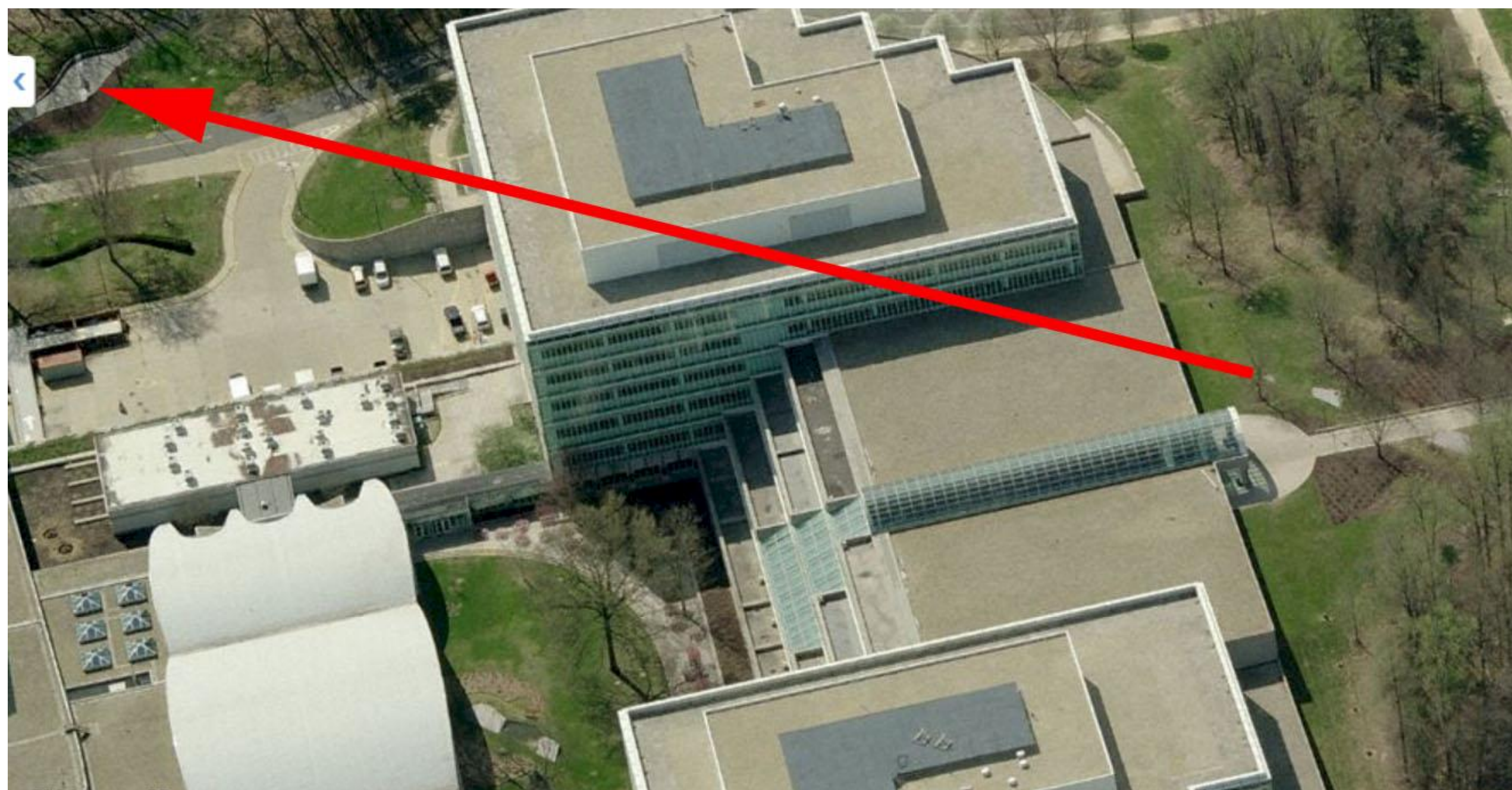
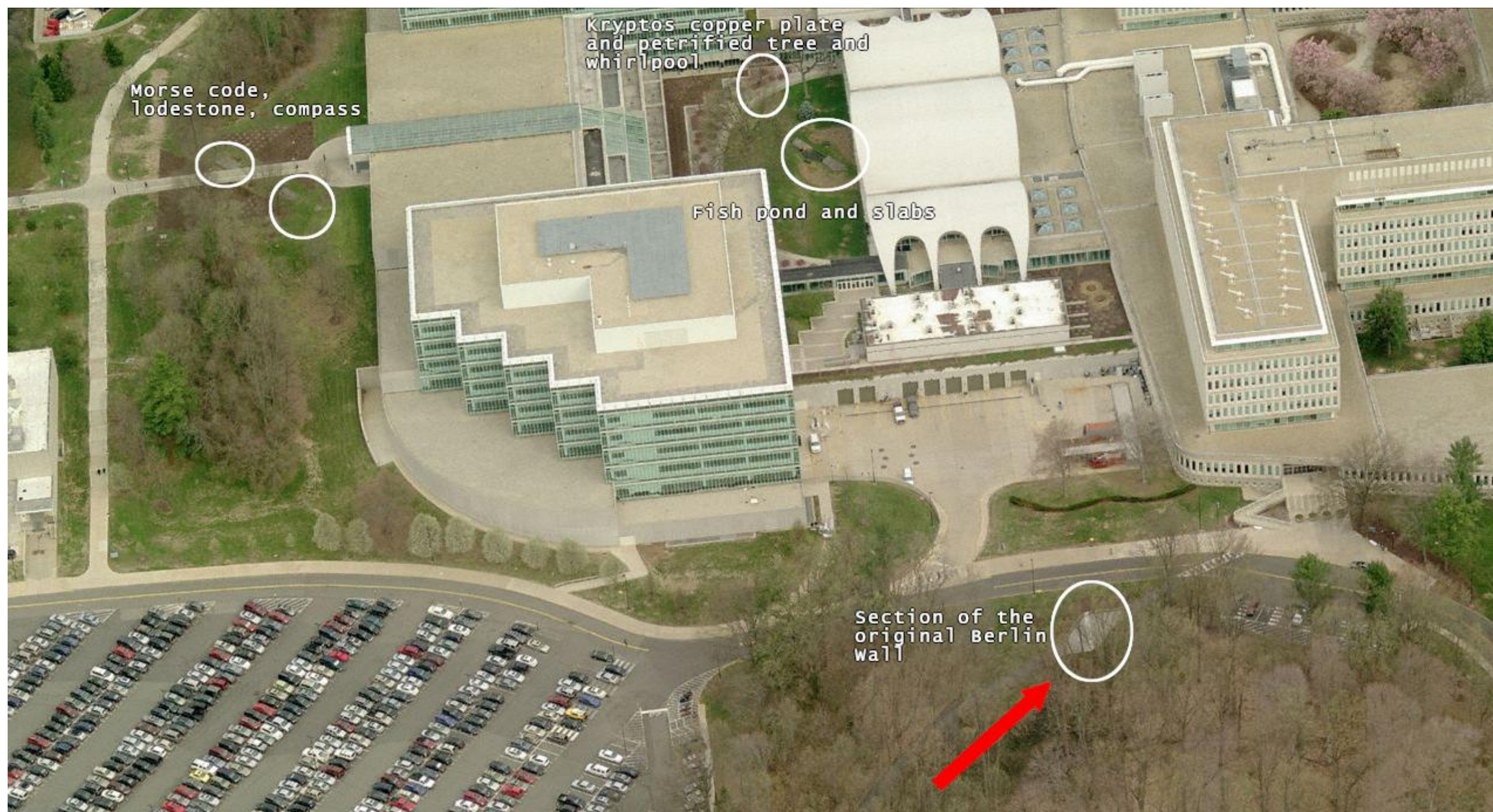


The Berlin wall

Jim Sanborn was given the commission to create Kryptos in 1988. It was formally dedicated on November 3, 1990. In between the Berlin Wall was first breached on 9 November 1989 and the wall came down. Jim Sanborn has said he was influenced by the wall coming down just like other people were but has never hinted that K4 is connected to the Berlin Wall. In November 2010 Jim Sanborn released a clue stating the letters NYPVTT in K4 were BERLIN (or NILREB). This has added to speculation that K4 may be connected to the Berlin Wall.

On CIA grounds there is a monument to the Berlin wall, an actual 3 part section of the wall from Berlin. It is situated in the middle of a path. It was dedicated in December 1992. It is not part of the Kryptos art installation. A bronze plaque near the wall reads, "These three sections of reinforced concrete were removed from the Berlin Wall near Checkpoint Charlie at Potsdamer Platz in November 1989."





End of photos

Kryptos K1 to K4 and the Morse Code and how K1, K2 and K3 were solved



Jim Sanborn



Edward Scheidt



William Webster

After Jim Sanborn decided he wanted to incorporate some encrypted messages in his artwork, he was teamed with Ed Scheidt, who was in the process of retiring as the Chairman of the Central Intelligence Agency Cryptographic Center . Up until that point, Jim Sanborn had never used encryption or text in his work. Ed Scheidt taught various encryption methods to Jim Sanborn, who chose the exact messages to be encrypted. The solutions to Kryptos were placed in a sealed envelope and handed to William Webster who was at the time the Director of the Central Intelligence Agency. They were never checked by anyone other than Jim Sanborn.

The encryption was all done by hand by Jim Sanborn using paper and pencil and the concept was for a CIA field operative returning home to CIA to use just paper and a pencil to solve Kryptos. The encryption was completed while on a trip to retrieve the petrified wood for the sculpture. A photo of the original encoding charts for K1, K2 and K3 can be found at the end of this file. In the early days Jim Sanborn said K1, K2 and K3 could be solved by a field operative using pen and paper in a couple of days and K4 would take a little longer. Jim Sanborn has said that you do not need to solve K1, K2 and K3 to solve K4 but has also said there are clues in K1, K2 and K3 to solve K4. He has said all clues are in plain sight. Ed Scheidt said "don't hold your breath" when referring to K4, possibly the answer is in one of the ponds.

California computer scientist Jim Gillogly was the first person in 1999 to announce the solutions to K1, K2 and K3. He used a Pentium II computer to do it. Then came an announcement (CIA had kept it secret) that a year earlier CIA analyst David Stein decrypted K1, K2 and K3 after spending some 400 hours worth of lunch hours working through the puzzles using only paper and pencil. Then came an announcement that the National Security Agency (NSA) had solved the first 3 parts in 1992. More than 20 years later no one has published the solution to K4.

KRYPTOS	
Cipher side	Tableau side
EMUFPHZLR FAXYUSDJKZLDKRN SHGNFIVJ YQTQUXQBQVYUVLLTREVJYQTMKYRDMFD VFPJUDEEHZWETZYVGWHKKQETGFQJNCE GGWHKK?DQMC PFQZDQMMIAGPFXHQRLG TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA QZGZLECGYUXUEENJTBJLBQCRTBJDFHRR YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE EVLDKFEZMOQQJLTTUGSYQPFEUNLAVIDX FLGGTEZ?FKZBSFDQVGOGIPUFXHHDRKF FHQNTGPUAECNUVPDJMQCLQUMUNEDFQ ELZZVRRGKFFVOEEXBDMVPNFQXEZLGRE DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG	ABCDEFGHIJKLMN O PQRSTU VWXYZ ABCD AKRYPTOSABCDEFGHIJLMNQUVWXZK RYP BRYPTOSABCDEFGHIJLMNQUVWXZK RYPT CYPTOSABCDEFGHIJLMNQUVWXZK RYPTO DPTOSABCDEFGHIJLMNQUVWXZK RYPTOS ETOSABCDEFGHIJLMNQUVWXZK RYPTOSA FOSABCDEFGHIJLMNQUVWXZK RYPTOSAB GSABCDEFGHIJLMNQUVWXZK RYPTOSABC HABCDEFGHIJLMNQUVWXZK RYPTOSABCD IBCDFGHIJLMNQUVWXZK RYPTOSABCDE JCDEFGHIJLMNQUVWXZK RYPTOSABCDEF KDEFGHIJLMNQUVWXZK RYPTOSABCDEFG LEFGHIJLMNQUVWXZK RYPTOSABCDEFGH MFGHIJLMNQUVWXZK RYPTOSABCDEFGHI
EN DY ^{AH} R OHNLSRHEOCPT EOIBIDYSHNAIA CHTNREYULDSLLSLLNOHSNOSMRWXMNE TPRNGATIHNRRAPESLNNELEBLPIIACAE WMTWNDITEENRAHCTENEUDRETNHAE OE TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR EIFTBRSPAMHHEWENATAMATEGYEERLB TEEFOASFIOTUETUAEOTOARMAEERTNRTI BSEDDNIAAHTTMSTEWPIEROAGRIEWFE B AECTDDHILCEIHSITEGOEAO SDDRYDLORIT RKLML E HAGTDHARDPNEOHMGFMFEUHE ECDMRIPFEIMEHNLSSTTRTVDOHW?OBKR UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO TWTQSJQSSEKZZWATJKLUDIAWINFBNYP VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR	NGHIJLMNQUVWXZK RYPTOSABCDEFGHIJL OHIJLMNQUVWXZK RYPTOSABCDEFGHIJL PIJLMNQUVWXZK RYPTOSABCDEFGHIJLM QJLMNQUVWXZK RYPTOSABCDEFGHIJLMN RLMNQUVWXZK RYPTOSABCDEFGHIJLMNQ SMNQUVWXZK RYPTOSABCDEFGHIJLMNQU TNQUVWXZK RYPTOSABCDEFGHIJLMNQUV UQUVWXZK RYPTOSABCDEFGHIJLMNQUVW VUVWXZK RYPTOSABCDEFGHIJLMNQUVWX WVWXZK RYPTOSABCDEFGHIJLMNQUVWXZ XWXZK RYPTOSABCDEFGHIJLMNQUVWXZK YXZK RYPTOSABCDEFGHIJLMNQUVWXZKR ZZK RYPTOSABCDEFGHIJLMNQUVWXZKRY ABCDEFGHIJKLMN O PQRSTU VWXYZ ABCD

The letters YAR on the first line of the bottom cipher text box are raised on the sculpture also.



A simple way to describe classical ciphers (used by K1 to K3)

You either use a scheme to change the actual letters or use a scheme to change where they are in the text.

"Substitution" ciphers substitute the letters in the alphabet for other letters. One version of this is to reverse the alphabet, so that when you want to write ABCDE you instead write ZYXWV.

"Transposition" ciphers do not change any letters in the message, they re-arrange them in a format that can be hard to work out if you don't know how. ABCDE could be DACBE and it could also be EABCD.

K1

The first 63 letters in the top left box are K1. They are " EMUFPHZLRFAXYUSDJKZLDKRNSHGNFIVJYQTQUXQBQVYUVLLTREVJYQTMKYRDMFD "

K1 uses a modified Vigenère cipher. It's encrypted through substitution – each letter corresponds to another– and can be solved only with the alphabetic rows of letters on the right on the Tableau side of the sculpture. The keywords, which help determine the substitutions, are KRYPTOS and PALIMPSEST. A misspelling – in this case IQLUSION – may be a clue to cracking K4.

K1 solution (13 words):

BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF IQLUSION

K2

The next 369 letters and 3 question marks (?) in the 12 lines under K1 are K2. They are "

VFPJUDEEHZWETZYVGWHKKQETGFQJNCE
GGWHKK?DQMCPFQZDQMMIAGPFXHQRLG
TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA
QZGZLECGYUXUEENJTBJLBQCRTBJDFHRR
YIZETKZEMVDUFKSJHKFHHKUWQLSZFTI
HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE
EVLDKFEZMOQQJLTTUGSYQPFEUNLAVIDX
FLGGTEZ?FKZBSFDQVGOGIPUFXXHHRKF
FHQNTGPPUAECNUVPDJMQCLQUMUNEDFQ
ELZZVRRGKFFVOEEXBDMVPMNFQXEZLGRE
DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP
DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG "

K2, like the first section, was also encrypted using the alphabets on the right Tableau side. One new trick Jim Sanborn used, though, was to insert an X between some sentences, making it harder to crack the code by tabulating letter frequency. The keywords here are KRYPTOS and ABSCISSA. And there's another intriguing misspelling: UNDERGRUUND.

K2 solution (75 words and 3 question marks (?)):

IT WAS TOTALLY INVISIBLE HOWS THAT POSSIBLE ? THEY USED THE EARTHS MAGNETIC FIELD X THE INFORMATION WAS GATHERED AND TRANSMITTED
UNDERGRUUND TO AN UNKNOWN LOCATION X DOES LANGLEY KNOW ABOUT THIS ? THEY SHOULD ITS BURIED OUT THERE SOMEWHERE X WHO KNOWS THE
EXACT LOCATION ? ONLY W_W THIS WAS HIS LAST MESSAGE X THIRTY EIGHT DEGREES FIFTY SEVEN MINUTES SIX POINT FIVE SECONDS NORTH SEVENTY SEVEN
DEGREES EIGHT MINUTES FORTY FOUR SECONDS WEST X LAYER TWO

K2 originally ended "WEST ID BY ROW_S" until in 2006 Jim Sanborn pointed out he had missed a letter out on the sculpture. Jim Sanborn has since said "w_w" refers to William Webster. The co-ordinates in K2, 38 57 6.5 N 77 8 44 W point to a location in the Courtyard on the CIA grounds. Jim Sanborn has said they could point to 2 locations. It is possible they were supposed to point to the sculpture.



K3

The next 336 letters in the 11 lines under k2 excluding the last 4 letters on the 11th line are k3. They are "

EN DYahr OHnLSRHEOCPTeOIBIDYSHNAIA
CHtNREYULDSLLSLLNOHSNOSMRWXMNE
TPrNGATIHNrARPESLNNELEBLPIIACAE
WMTWNDITEENRAHCTENEUDRETNHAEOE
TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR
EIFTBRSPAMHHEWENATAMATEGYEERLB
TEEFoASFIOTUETUAEOTOARMAEERTNRTI
BSedDNIAAHTTMSTEWPIEROAGRIEFEB
AECTDDHILCEIHSITEGOEAOSDDRYDLORIT
RKLMLEHAGTDHARDPNEOHMGFMFEUHE
ECDMRIPFEIMEHNLSSTTRTVDOHW "

Jim Sanborn has said the ? is the start of k4.

A different technique was used for K3: transposition. All the letters are jumbled and can be deciphered only by uncovering the complex matrices and mathematics that determined their misplacement. Of course, there is a misspelling (DESPARATLY), and the last sentence (CAN YOU SEE ANYTHING?) is strangely bracketed by an X and a Q.

K3 Solution (75 words) :

SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER AND THEN WIDENING THE HOLE A LITTLE I INSERTED THE CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q (?)

K3 is a paraphrased quotation from Howard Carter's account of the opening of the tomb of Tutankhamun on November 26, 1922, as described in his 1923 book The Tomb of Tutankhamun. The question with which it ends is that posed by Lord Carnarvon, to which Carter (in the book) famously replied "wonderful things". In the actual November 26, 1922 field notes, his reply was, "Yes, it is wonderful.". The Tomb of Tutankhamun was one of Jim Sanborn's favourite books and he has said that solving Kryptos could be similar to entering the tomb, with one doorway leading to another.

There are a few different ways K3 can be solved, the simplest way (not how Jim Sanborn intended) is to rotate K3 right by 90 degrees with a column width of 24. This is the result:

DAEGIFWQEWRNDEMGGRBIEHONAOYNRTOI
SONEEDRHSDIDEEETAETIPSHyPHAWDU
TNFTENNaFAOFDEACOESOAHERSENTMTL
ELSiRIDDBIUAAASNMAOMPDAATYERN
RCHENREAEECDAEWHNHEYCHOGRTHLXLN
ODTTTYEICEMNSLHLDTOEIWTBNRRSMO
DMAEFEELEEHSGRHSRRTNPTYETFIITML
BHEIPUOTMTLEABRAUIRLCRFRCWETSED
ANDPTEKEPEEPIRCGSTVULIIREAOEAALED
HMHETfMYTETLOOELSRNOHTNWISIHE
EIORAHEHMHlBWCHTATSEYATNLI

If you apply the same technique on the result but with a column width of 8 you find the answer to K3.

K4

The question mark '?' and the last 97 letters are K4. They are "

?OBKR
UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO
TWTQSJQSSEKZZWATJKLUDIAWINFBNYP
VTTMZFPKWGDkZXTJCDIGKUHUAUEKCAR "

Although K3 ends with a question and may also use the ? Jim Sanborn has said K4 starts with the ?

No solution has ever been published. It is not known if the CIA or NSA have solved it. They never revealed they had solved the first 3 parts until the solutions were made public. Many Freedom of Information requests have been made to CIA and NSA and many documents received in response but never the answer to K4. NSA have stated that when they looked at K4 and decided what type of encoding was most likely to have been used they decided not to proceed. That's the official story.

One reason that K4 has proven difficult to crack is that according to Jim Sanborn and Ed Scheidt a masking technique was used. In addition to obtaining a ciphertext for K4 Jim Sanborn used ways to hide what that the plaintext or ciphertext should look like. This had made it impossible, as far as we know, for even computers to solve it. Masking hides the ciphertext. Removing all vowels could be used, deleting letters, changing every nth letter to the one before it in the alphabet are all forms of masking. There are many ways to mask the plaintext and the ciphertext. Ed Scheidt said to crack K4 you need to work out the techniques used first.

The Morse Code

On the slabs on the front lawn the following morse code is cut out in the copper sheets.

... .. |

ee V I R T U A L L Y e | e e e e e I N V I S I B L E

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-.. .. --. . - .- .:.. . . . | .. -. - . .: .--. .:.. . - .- - .. -cut

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D I G E T A L e e e | I N T E R P R E T A T I - ?

or

D I G e T A L E e e | I N T E R P R E T A T I - ?

. |

ee S H A D O W ee | F O R C E S ee eee

..... |

L U C I D e e e | M E M O R Y e

— —, — — — .. —, —, | —, — — — .. — — — —, —

T I S Y O U R | P O S I T I O N (Some people think this says "what is your position")

...

S O S (SOS is the international Morse code distress signal. The Titanic used the older CQD)

• — • — • —

R Q (Is used in Morse Code to indicate a question)

Jim Sanborn's Original Encoding Charts For K1, K2 and K3

[illegible]

ABSCISSA

3	I	T	W	A	S	T	O	T	A	L	L	V	I	N	V	I	S	I	B	L	E	H	O	W	S	T	H	A	T	P	O	
	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	
	V	F	P	J	U	D	E	E	H	Z	W	E	T	Z	Y	V	G	W	H	K	K	Q	E	T	G	F	Q	J	V	C	E	
4	S	S	I	B	L	E	?	T	H	E	Y	U	S	E	D	T	H	E	E	A	R	T	H	S	M	A	G	N	E	T	I	
	A	A	B	S	C	I	?	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	
	G	G	W	H	K	K	?	D	Q	M	C	P	F	Q	Z	D	Q	M	M	I	A	G	P	F	X	H	Q	R	L	G	T	
5	C	F	I	E	L	O	X	T	H	E	I	N	F	O	R	M	A	T	I	O	N	W	A	S	G	A	T	H	E	R	E	D
	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I
	I	M	V	M	Z	J	A	N	Q	L	V	K	Q	E	D	A	G	D	V	F	R	P	J	U	N	G	E	U	N	A	Q	Z
6	A	N	D	T	R	A	N	S	M	I	T	T	E	O	U	N	D	E	R	G	R	O	U	N	D	T	O	A	N	U	N	
	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	
	G	Z	L	E	C	G	Y	U	X	U	E	E	N	J	T	B	J	L	B	Q	C	E	T	B	J	D	F	H	R	R	Y	
7	K	N	O	W	N	L	O	C	A	T	I	O	N	X	D	O	E	S	L	A	N	G	L	E	Y	K	N	O	W	A	B	
	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	
	I	Z	E	T	K	Z	E	M	V	D	U	F	K	S	J	H	K	F	W	H	K	U	W	Q	L	S	Z	F	T	I	H	
8	O	U	T	T	H	I	S	?	T	H	E	Y	S	H	O	U	L	D	I	T	S	B	U	R	I	E	D	O	U	T	T	
	C	I	S	S	A	A	B	?	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	B	S	C	I	S	S	A	A	
	H	D	D	D	U	V	H	?	D	W	K	B	F	U	F	P	W	N	T	D	F	I	Y	C	U	Q	Z	E	R	E	E	

Jim Sanborn has said examining the charts forensically may provide clues

Jim Sanborn's Letter to CIA Employees

C00042168

December 15, 1989

Dear Agency Employees:

I am writing this letter to give you an idea of what I am up to at the Agency, and to explain those big tilted slabs of stone.

The stonework in the courtyard and at the entrance to the new building serves two functions.

First, it creates a natural framework for the project as a whole and is part of a landscaping scheme designed to recall the natural stone outcroppings that existed on this site before the Agency, and that will endure as do mountains.

Second, the tilted strata tell a story like pages of a document. Over the next several months, a flat copper sheet through which letters and symbols are cut will be inserted between these stone "pages." This code, which includes certain ancient ciphers, begins as International Morse and increases in complexity as you move through the piece at the entrance and into the courtyard. Its placement in a geologic context reinforces the text's "hiddenness" as if it were a fossil or an image frozen in time.

An installation in the courtyard further explores this theme. On the paved surface, supported by a petrified tree, will stand a curved, vertical copper plate. Approximately 2000 letters of the alphabet are cut through this plate (a process which requires four months of work). The left side of the plate is a table for deciphering and enciphering code, developed by Blaise de Vigenere in 1570.

Approved for Release
Date JAN 1993

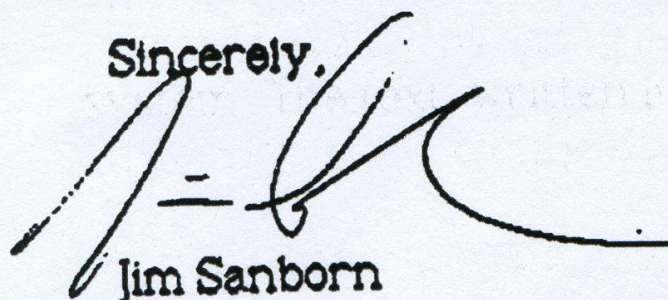
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The right side is a text that can be partly deciphered by using the table and partly by using a potentially challenging encoding system. The text, written in collaboration with a prominent fiction writer, is revealed only after the code is deciphered.

My choice of materials, like code, conveys meaning. At the entrance, a lodestone (a rock naturally magnetized by lighting) refers to ancient navigational compasses. The petrified tree recalls the trees that once stood on this site and that were the source of materials on which written language has been recorded. The copper, perforated by text, represents this "paper." I also use another symbol; water. In a small pool on the plaza, partly surrounded by the copper plate, water will be turbulent and provocative, constantly agitated into standing waves. In the other pool, located among trees in the courtyard and between two massive outcroppings, water will be calm, reflective, contemplative. Other materials around the site - large stones, ornamental grasses, and small trees - are designed to make the natural features surrounding the Agency more visually interesting and thought provoking.

My work at Langley is approximately two thirds complete. If you see me or my apprentices working, please don't hesitate to ask questions about the work.

Sincerely,



Jim Sanborn